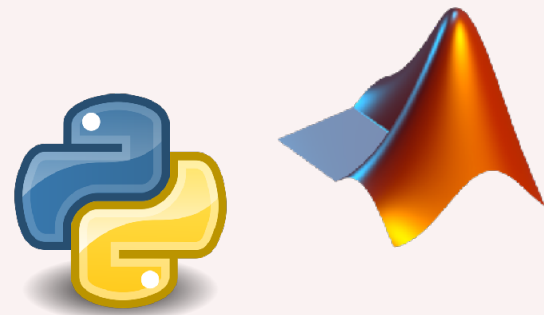


Introduction to Julia for High-Performance Computing

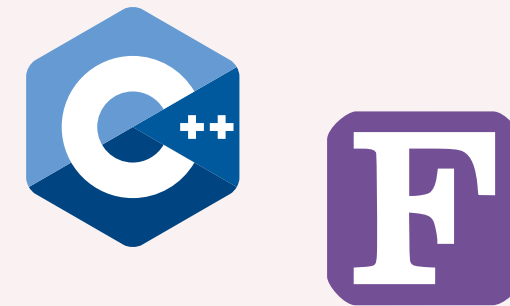
Carsten Bauer @ HLRS, Stuttgart

September 23, 2025

Convenience

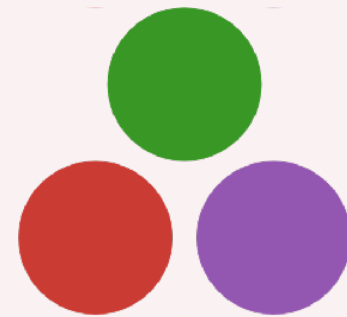


Performance



Language Barrier

Convenience



Performance



Gradual transition

	Tuesday	Wednesday	Thursday	Friday
	Foundations	Core	Node	Cluster
09:00 - 10:45	Intro Onboarding	Type & Memory Optimizations	Multithreading	Distributed Computing
10:45 - 11:00	Break	Break	Break	Break
11:00 - 12:30	Fundamentals	Exercises	Exercises	Exercises
12:30 - 14:00	Lunch	Lunch	Lunch	Lunch
14:00 - 15:30	Compilation	SIMD	GPU Computing	Profiling
15:30 - 15:45	Break	Break	Break	Outro
15:45 - 17:00	Exercises	Exercises	Exercises	

Quick Live Survey

Julia's Weaknesses

HPC with Julia is
currently a **niche**.

Join us at conferences ...



... or in our monthly Zoom call
(open to everyone!)

No great way to
produce (small)
binaries.

Achieving
high performance
can be tricky.

Julia's Strengths

Julia is **interactive**
and **convenient**.

Julia has a great package manager

Laptop



```
→ ~/myproject tree
.
├── Manifest.toml
├── Project.toml
└── code.jl

0 directories, 3 files

→ ~/myproject cat Project.toml
[deps]
CUDA = "052768ef-5323-5732-b1bb-66c8b64840ba"
DifferentialEquations = "0c46a032-eb83-5123-abaf-570d42b7fbaa"
MKL = "33e6dc65-8f57-5167-99aa-e5a354878fb2"
MPI = "da04e1cc-30fd-572f-bb4f-1f8673147195"

→ ~/myproject
```

HPC Cluster



```
→ bauer@n2login3 myproject julia --project
```

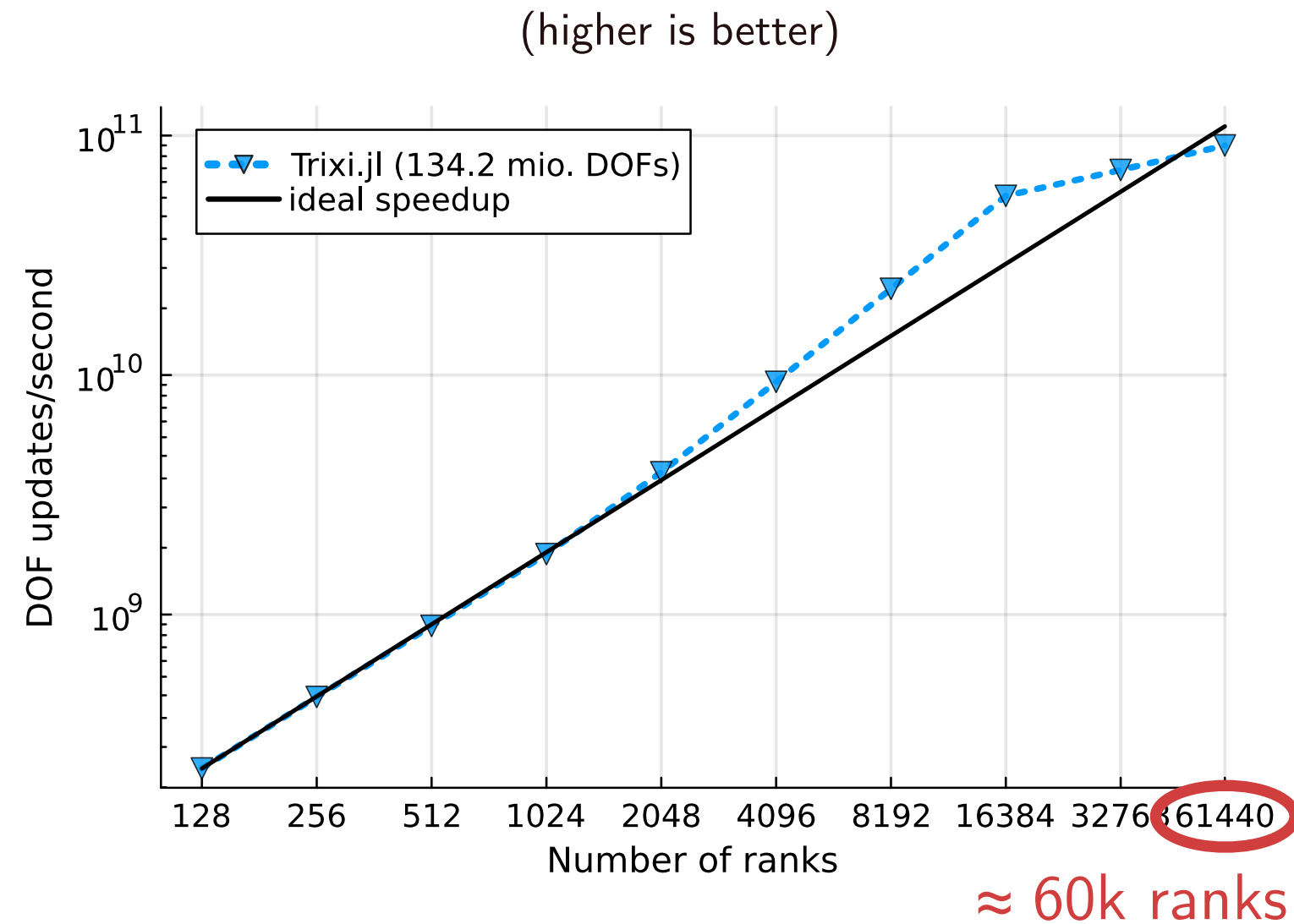
```
(-) | (-) |  
(-) | (-) | (-)  
| | | | | | / \ |  
_/_\_|\_--|\_--|_-|\_--|_  
|--/
```

```
Documentation: https://docs.julia.org  
Type "?" for help, "]"?" for Pkg help.  
Version 1.7.2 (2022-02-06)  
Official https://julia.org/ release
```

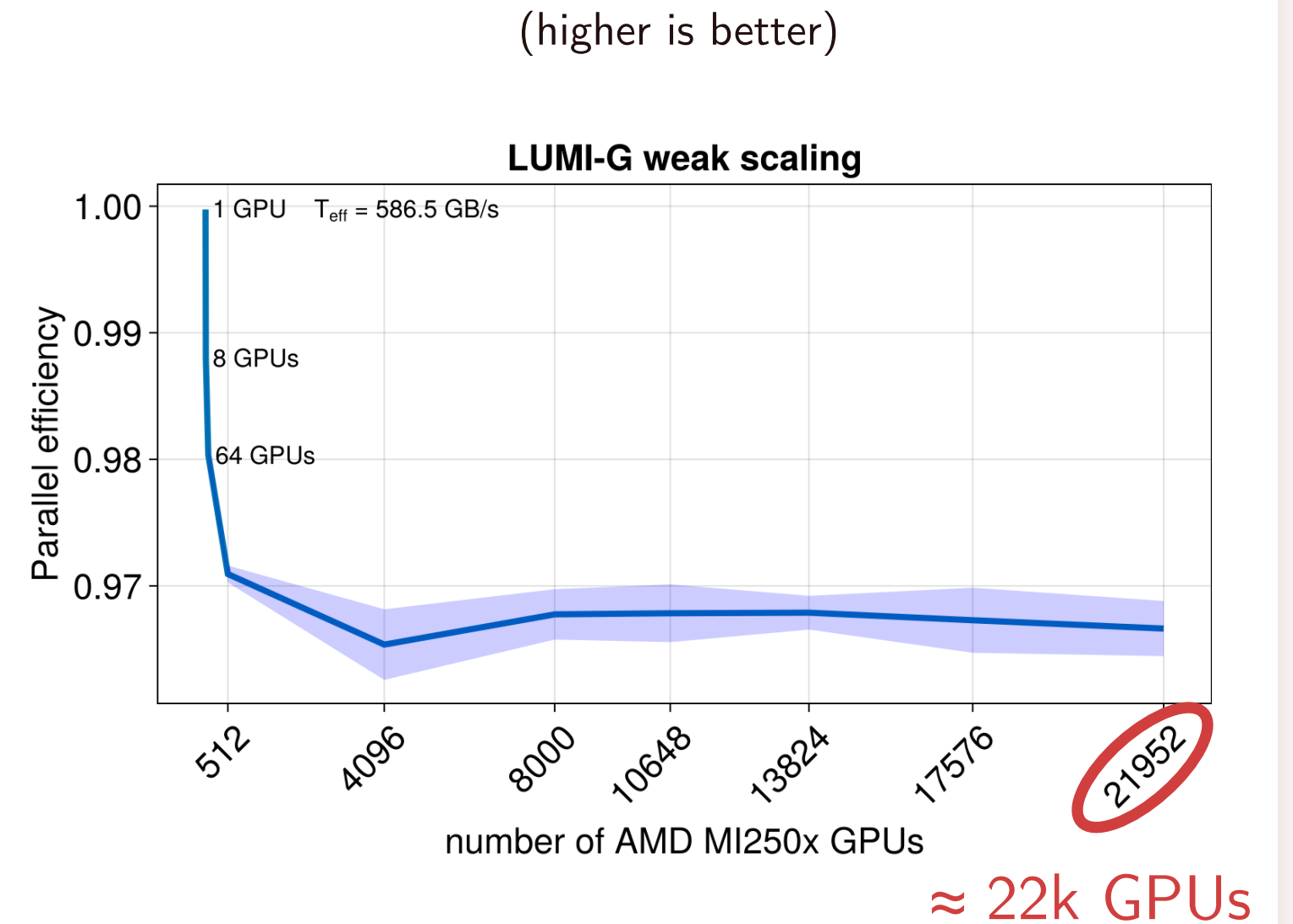
```
(myproject) pkg> st  
Status `~/myproject/Project.toml`  
→ [052768ef] CUDA v3.11.0  
→ [0c46a032] DifferentialEquations v7.1.0  
→ [33e6dc65] MKL v0.5.0  
→ [da04e1cc] MPI v0.19.2  
Info packages marked with → not downloaded, use `instantiate`  
to download  
  
(myproject) pkg> instantiate
```

Julia code can be
fast and **scalable**.

Example: Good scaling of PDE codes



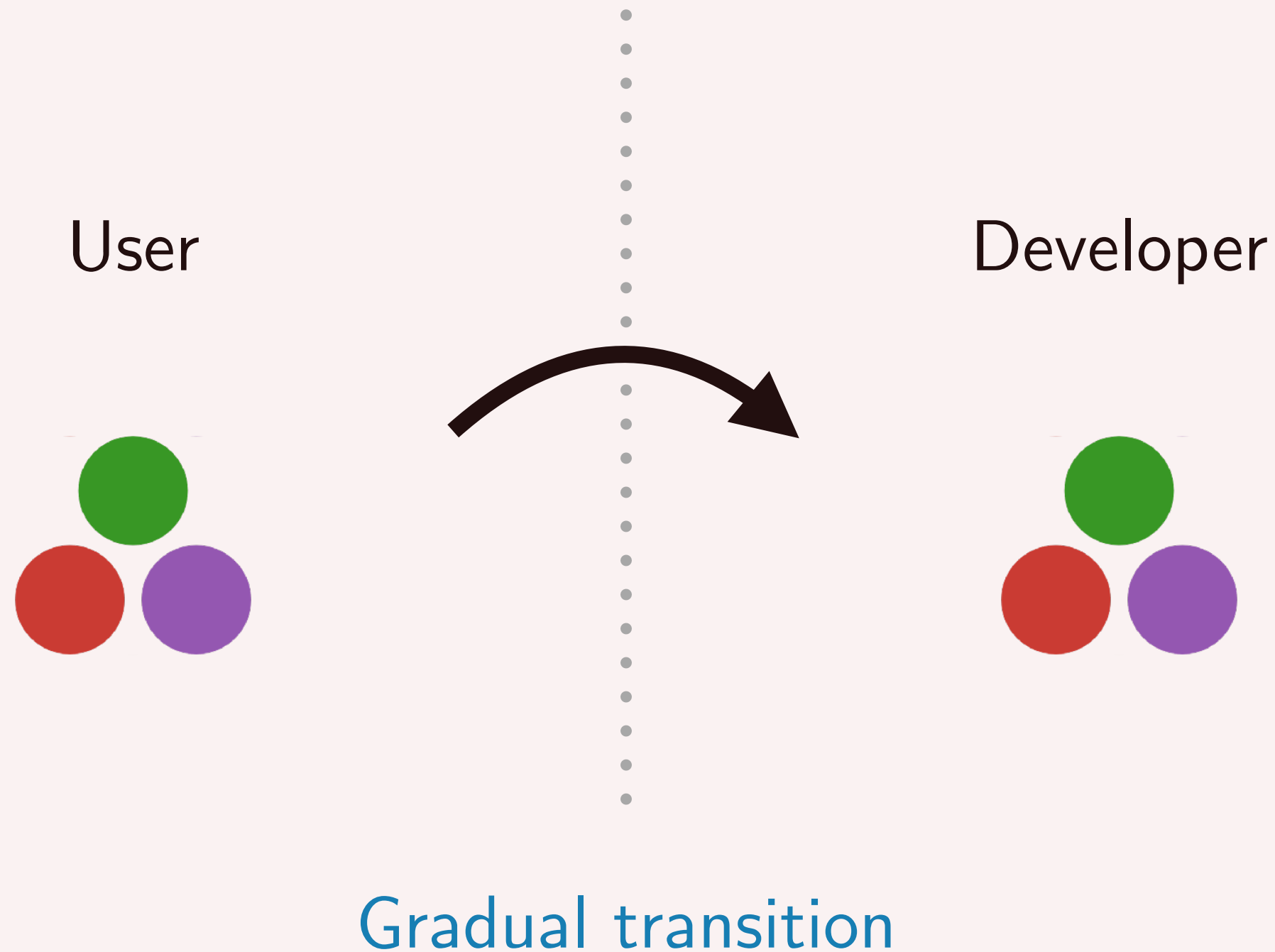
Trixi.jl (Multi-CPU)



ParallelStencil.jl (Multi-GPU)

Julia invites you to
gradually **delve deeper.**

Julia makes it easier to become a developer.



Let us delve deeper!

